

Institute of Actuaries of India

Subject ST1 – Health and Care Insurance Specialist Technical

April 2016 Examinations

INDICATIVE SOLUTIONS

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i)

- Underwriting can protect an insurance company from anti-selection
-and in particular from lives whose health is so seriously impaired that it is impossible to assess the risk accurately.
- It enables insurers to identify lives with a substandard health risk for whom special terms must be quoted.
- For the substandard risks, it will identify the most suitable approach and premium level for the special terms to be offered.
- Adequate risk classification within the underwriting process will help to ensure that all risks are rated fairly.
- Underwriting will help in ensuring that actual morbidity experience does not depart too far from that assumed in the pricing of the contracts being sold.
- It may help to obtain reinsurance protection and at affordable rates.
- Financial underwriting will help to reduce the risk from over insurance.
- Claims stage underwriting ensures only valid claims are paid.

[4]

ii)

The possible rating factors are:

- Individual product
 - Age
 - Gender
 - Smoker Status
 - Location of residence
 - Deferred period
 - Linked claim period
 - Distributional channel
 - Income
 - Occupation
 - Premium payment frequency
- Group Product, additionally,
 - Size of the group (no. of members)
 - Type of group (Employer-employee or Affinity)
 - Industry
 - Level of participation (compulsory vs voluntary)

[3]

iii)

The reasons for high operating costs may be:

General

- The increase may be in line with the general level of inflation, which has been high recently
- The increase might reflect high wage inflation e.g. because of competition for qualified/trained staff due to new companies being set up
- Higher staff costs due to new labor laws increasing minimum wage, lowering working hours, increasing social security contributions etc.
- High staff turn-over leading to higher recruitment and training costs and lower efficiency
- Increased staff costs due to significant retention bonus in order to retain key staff, increase in personal income tax forcing the company to increase salaries etc.
- If the increase identified is the cost per policy, then it may have been due to a fall in new business levels, so that fixed costs have been spread over fewer policies
- Cost of IT systems might have increased (eg. more sophisticated IT infrastructure for cyber security)
- Increased policy servicing costs due to change in regulatory compliance requirements (eg. more frequent and detailed communication to the policyholders, additional levels of scrutiny before a claim is repudiated etc.)
- Increased legal costs due to policyholder complaints, more comprehensive/stringent legal process with respect to repudiation of claims,
- increased fraud incidences leading to more investigations etc.
- Introduction of a new distribution channel (eg. Agency) which is infrastructure heavy and so the per policy costs have increased
- Increased property costs (eg. shifting of office premises to higher cost location, improved building security procedures)
- Change in corporate governance requirements leading to additional staff and infrastructure
- Increased regulatory compliance cost due to change in regulations (eg. increased frequency of audits, more detailed process documentation, requirement to issue sales literature and policy documents in multiple languages etc.)

Underwriting

- Increased underwriting costs due to medical inflation, which might have been exceptionally high recently e.g. the introduction of more costly and sophisticated special medical tests.
- The insurer might have grown its new business volumes significantly, and hence there has been a significant increase in total cost of underwriting, although the cost of underwriting per policy might not have increased significantly.
- There might be a different customer base which results in an increase in the average per policy cost of underwriting e.g. due to using a new distribution channel, or a shift from internet distribution to broker distribution,
- or a shift from group business to individual business.

- There may have been a shift towards older customers, triggering earlier the non-underwriting limits, or customers of worse health, requiring extra investigations to find appropriate policy loadings or larger policy sizes, triggering more evidence limits, or customers in different territories where the cost of underwriting is higher.
- For group business, it could be due to lowering of free cover limits or high levels being bought leading more cases being individually underwritten.
- There could be a shift in product mix if different levels of underwriting are used.
- A reinsurer might have required the insurer to improve the stringency or quality of its underwriting or it might have decided to do this in order to obtain cheaper reinsurance rates.
- There may have been more claims underwriting because of an increase in the number of claims or changes in the claims definition (e.g. becoming more complex).
- There may have been an increase in claim fraud and nondisclosure, so increased claim underwriting has been needed (if this is included in the costs that are being considered).
- The insurer might have changed its underwriting strategy to be more stringent e.g. moved from medical history disregarded to full underwriting,
- or no longer using reinsurance and losing the technical expertise provided by the reinsurer.
- This may have increased the amount of information asked for on the proposal form and/or increased the cost due to a higher number of medical exams, doctors' reports. For example, it might have reduced the sum insured limits which trigger the additional evidence requirements.
- New underwriting strategies may have evolved (e.g. based on genetics).
- The insurer might have found that it needed to use more highly qualified and hence costly staff in its underwriting team in order to maintain standards.
- The insurer might have decided to reduce its use of pre-existing condition exclusions and hence needs to perform more underwriting in order to assess additional premium loadings. Similarly, the insurer might have decided to reduce the number of declinatures at outset.
- If the increase relates to initial underwriting costs rather than claims management costs, then it may reflect a change in focus from the latter to the former e.g. due to poor publicity following refused claims.
- There may have been changes in regulatory/compliance requirements, with related underwriting costs e.g. more documentation.
- The insurer might have invested in new underwriting system to improve long term efficiencies.

[12]

iv)

- Full medical underwriting to SMQ based underwriting for Individual
 - the disability inception rates will likely increase as there is now more scope for anti-selection
 - this would, however, depend upon
 - the distribution channel (for example, agency channel tends to be more anti-selective than the banca), and
 - whether the insurer has access to any other proxies (eg. family doctor's reports, health check-up reports etc.), and

- length of waiting period and deferred period (the longer the waiting period the greater the negative impact will be on the experience), and
 - whether the applicant pool is pre-selected (for example, the product is targeted to a pre-selected pool of lives with some proxy indicators of current health)
- it also depends on the type of questions asked and whether it would be 'accept' or 'reject' based on the SMQ responses or there would be any follow up questionnaire or medical if there is any adverse medical history disclosed in the SMQ
- the termination rates largely depend upon the claim underwriting and so the termination experience will likely not be impacted materially
- No individual member level underwriting for Group
 - The impact on disability inception experience would depend upon the type of participation (the experience would deteriorate more heavily for voluntary participation than for compulsory participation), and
- Type of cover structure (the level of cover being very high for some members relative to others might encourage anti-selective behavior and thus result in heavier experience)
- Compulsory groups that have flat cover structure (same level of cover for all members) at a reasonable level would any way not typically require individual member-wise underwriting and so no material impact due to the change in underwriting approach
- Different types of groups will drive different anti-selective behavior and thus impacting the experience differently (employer-employee groups will be less anti-selective compared to certain affinity groups and so the impact on experience will be relatively lower)
- the impact will also depend upon how the competitors are underwriting (if the competition does only full medical underwriting, then there will likely be heavier impact due to greater anti-selection potential)

[6]

[25 Marks]**Solution 2:**

- i) The persistency experience should be monitored for:
 - The analysis shall be used for setting assumption to calculate profitability of the product and business planning
 - The actual renewal experience shall be compared with the renewal assumption used at the time of pricing. Monitor the reason for variances.
 - For example: monitoring persistency by distribution channel.
 - This is significant as the initial costs are amortized over a number of years of renewal and any adverse experience will affect the recovery of initial expenses and hence the profitability
 - Having an appropriate renewal rate assumption for PMI is essential in order to amortize the initial expense costs when pricing the product.
 - The analysis will also enable the company to identify groups, eg particular agents, channels or regions that may have very different renewal experience.

- The company can then take appropriate remedial action, eg change its sales distribution and/or marketing strategies.
- Renewal rates, split according to previous claims history can be used to set appropriate NCD scales, where applicable.
- This type of analysis also helps in setting commission claw-back conditions
- Lower than expected persistency may also imply anti-selective withdrawals (healthy lives leaving and the poor health lives staying)
- ...and thus making the experience of continuing policyholders heavier than expected – so, the persistency analysis help taking business decisions as to sales focus and retention strategies

[4]

ii)

The first step is to analyse product's own experience in terms of:

- Claims Incidence Rate
- Claims Size
- Expenses
- Persistency

Claim Incidence Rate

- We will need to analyse the past experience.
- The data will be subdivided by:
 - benefit / procedure class
 - age
 - gender
 - smoker status
 - occupation
 - distribution source
 - territory / region.
- Several years of recent experience may be combined in order to produce credible data.
- But we must avoid introducing too much heterogeneity by keeping the period as short as is compatible with adequate data for credible estimates.
- Care needs to be exercised in terms of any change in claim definitions or philosophy or administration (eg. in-house vs TPA) when interpreting the results
- Numbers of claims would be divided by the number of observed policy years, to obtain the historical claim incidence rate (for each grouping).
- Trends in experience may be identified by analysing the data over time.
- External data may be used to support the internal investigation, eg:
 - reinsurers' data
 - industry data
 - market data
- However, differences in applicability and relevance of these data must be taken into account. Such data may also be useful for identifying trends.

- The historical incidence rates should then be projected over the next year (or number of years, if any guarantee is being offered on keeping rates unchanged on renewal or if the company is hoping to keep its rates constant over a period, eg. for marketing reasons).
- Changes that may affect claim incidence rates need to be taken into account when projecting the rates, eg changes in:
 - economic conditions
 - policy benefits provided
 - mix of business
 - policy conditions
 - types of treatments covered etc.
- Additionally, future rates may be adjusted to allow for higher incidence rates among policyholders who have claimed at least once before under their policies.
- The extent of any adjustment would be estimated from the past experience in this regard.

Claims Size

- Using past data subdivided as above, calculate average claim cost per claim ... by dividing the total claim costs by the number of claims involved.
- Care needs to be exercised in terms of any change in claim definitions or philosophy or administration (eg. in-house vs TPA) when interpreting the results
- Claim amounts will be projected to the period for which the new rates will apply ... allowing for the impact of changes in:
 - medical treatments
 - costs of treatments
 - medical protocols
 - hospital charging structures
 - inflation of medical costs
 - the mix of business (eg age, gender, distribution channel, occupation, policy size) where these are not fully allowed for in the pricing structure.

Expenses

- The company will need to analyse its recent expenses due to:
 - initial expenses of acquiring a new policy (selling, underwriting, admin)
 - expenses of renewing a policy
 - expenses of policy termination (ie of non-renewal)
 - claims management expenses
- Data may need to be divided into homogenous groups, if necessary to reflect unequal shares of costs, eg between distribution channels, gender, occupation.
- The expenses will be analysed relative to appropriate measures, eg:
 - selling expenses as a percentage of premium
 - claim and underwriting expenses as a percentage of average claim amounts
 - the remaining expenses as per-policy
- Dividing by the appropriate measures will give the historical expense loadings for the contract.

- The past expenses (and loading) will need to be split according to whether they were direct or overhead (fixed).
- The loadings will then need to be increased for inflation to the middle of the period for which they are expected to be appropriate.
- In order to estimate its expense inflation rate, the company would compare its historical expense increases against past changes in prices and earnings indices; expectations of the relevant index-inflation rates will then help determine the rate to use.
- An allowance will be made for spreading the initial expenses over the expected number of renewals of the contract – using the appropriate renewal rate assumptions.
- Commission would be allowed at the rate(s) that the company intends to pay.
- The additional cost of initial commission would be spread over the expected renewals in the same way as for other initial expenses.
- The appropriate per-policy expense assumptions would be obtained by considering total expected future new business volumes, including both new business and renewals, to work out each policy's expected contribution.

Persistency

- Numbers of renewals will be divided by the numbers of policies coming up for renewal, in the past experience data, as a starting point for estimating future renewal rates.
- Rates may be analysed separately by:
 - distribution channel
 - occupation
 - territory
 - duration since the policy was first sold (ie by the number of previous renewals)
- Any special conditions (eg economic circumstances or competitor activities) affecting the past experience should be taken into account before using the historical data to estimate the future experience
- The projected rates would also take account of:
 - changes in the (perceived) service delivery from State-provided alternatives
 - changes in premium rates due to inflation (eg a large increase in premium rates would discourage renewals)
 - changes in the economic outlook
 - Change in target market
 - the effect of the proposed change in premium rates, compared to the competition.

Calculating the risk premium

- The premium to cover the expected cost of claims is found by summing over all the different benefit classes covered by the policy.

$$\text{Risk premium (age, gender)} = \sum i_k \times A C_k$$

where i_k is the incidence rate for benefit / procedure class k
and $A C_k$ is the average claim cost for benefit / procedure class k .

ik and ACk are specific to each age and gender and the summation is over all classes k .

Calculating the office premium

- The office premium is obtained by adding the expense loadings to the appropriate risk premium for each risk group.
- The overall price may then need to be adjusted in relation to competitors' prices and the expected marketability of the product.
- The aim will be to maximise:

$$\{\text{Individual profit per policy}\} \times \{\text{number of new sales and renewals}\}$$
- Margins will need to be included in excess of best estimate assumptions, in order to introduce some risk protection and to generate an expected profit.

[18]

iii) The insurer could consider the following measures for improving market share without compromising on profitability:

- Increase the business volume so as to reduce the per policy costs through marketing campaigns in such a way that the benefits outweigh the cost significantly
- Centralizing and optimizing certain operational activities in a low cost location to reduce costs
- Outsourcing of certain activities (large outsourcing companies may offer attractive price due to their scale and operational efficiency) to reduce costs
- Outsourcing of claim administration to a TPA might result in cost savings as well as ensuring that only genuine claims are paid and thus improving claim experience
- Focus on less commission-intensive distribution channels (for example, direct to customer through internet, bancassurance, worksite marketing etc.)
- Simplify the underwriting and onboarding process to facilitate faster business acquisition and reduced cost
- Increase or decrease reinsurance depending upon the risk/return trade-off in order to reduce office premium rates without reducing profitability
- The guaranteed renewability for life, to the extent not perceived to be adding much value, can be reduced to, say, guaranteed renewability till age 60 – this might result in shareholders accepting a profit target lower than they would otherwise require
- Adding innovative product features that have much higher perceived value than the cost involved
- Customer engagement initiatives such as wellness programs in order to improve persistency and thus improving market share
- Offer a no claim discount to encourage good lives to stay on
- Acquire a competitor having good market share
- Offer more convenient ways of paying renewal premium (eg. online, ECS, etc.)

[6]

[28 Marks]

Solution 3:

- It is typical for insurance supervisors to require that an insurer maintains at least a specified level of solvency capital which is to provide additional level of protection to policyholders
- While it is true the supervisory reserves are provided to take care of the liabilities being met in given scenarios the additional solvency capital gives a cushion in case the experience turns out to be worse than that assumed in the reserve calculation
- This therefore enhances the customer confidence in the insurance company
- Adequacy of reserves is looked at in conjunction with solvency capital requirements and not in isolation. Similarly, adequacy of solvency capital is seen in conjunction with the method of determination of reserves
- The relative balance between the two varies depending upon the practices in different countries
- In some countries the reserves are set up on a realistic basis with smaller margins but then solvency capital are determined using risk based capital techniques i.e level of capital required is linked to the level of risk
- In other countries the reserves may be required to be maintained on a prudent basis with large margins and relatively small solvency capital, not necessarily linked to the inherent risks
- Even under the solvency II regime wherein the reserves are required to be calculated on a best estimate basis plus an additional margin, the insurer has to hold a solvency capital based on a 99.5% confidence interval that the value of assets should exceed liabilities over a one year time horizon
- Moreover the solvency capital does not remain idle. While they may be required to be invested in a prescribed manner and not according to the shareholder appetite, it is not going to remain fully unproductive. The prudent investments of assets backing solvency capital can actually be critical for the insurer in volatile markets.
- Therefore, solvency capital cannot be seen in isolation to reserve calculation and the two together are used to decide the solvency level of the company and ensure the protection of policyholders

[4 Marks]**Solution 4:****i) Proportional****Quota Share (QS)**

- QS is a proportional arrangement in which both premiums and claims are shared in a fixed proportion; for example a flat QS 50% would mean that 50% of the risk is passed on to the and the reinsurer will therefore pay 50% of every claim in return. The arrangement is very simple to administer.
- Since the risk is new to the insurer, a lower retention (say, 30% QS) can be considered subject to any regulatory restrictions on the level and type of reinsurance.

- The percentage QS can be varied for different levels of per day benefit if the insurer perceives different level of riskiness associated with different levels of per day benefit. For example, 40% for 1,000 per day cash and 70% for per day for 5,000 per day if the insurer feels 5,000 per day benefit is riskier than 1.000 per day benefit.
- The reinsurance premium can be on original terms basis (ie. a portion of the office premium is paid to the reinsurer and the insurer received a reinsurance commission) or, it can be on YRT risk premium basis (ie., the reinsurance premium will increase with age and no reinsurance commission payable).

Individual Surplus (IS)

- Again, IS is a proportional arrangement similar to QS except that the percentage of risk shared will vary depending upon the level of per day benefit. For example, retention of 2,000 per day benefit per policy would mean no reinsurance for policies with per day benefit lower or equal to 2,000. And, a $\frac{2}{5}$ th of a 5,000 per day benefit policy would be retained and $\frac{3}{5}$ th reinsured. This is a bit more complex to administer than a flat QS.
- IS for this product is, therefore, equal to QS with different percentage risk sharing for different per day benefits as outlined above

Non-Proportional

- Individual excess of loss – under this arrangement the reinsurer shares a portion of all claims under a policy in a given policy year in excess of the insurer's retention. For example, 50% in excess of 50,000 up to 300,000. If there is 15 weeks' of continuous hospitalization for a policy with per day benefit of 3,000 in a given policy year, the total claim amount would be 315,000 and the reinsurer would pay $50\% \times (315,000 - 50,000) = 132,500$
- Aggregate excess of loss – this is similar to the Individual excess of loss except that the claims would be aggregated over all policies for a given period of time (typically, one year) and reinsurer will be involved if the aggregate claim amount exceeds retention.
- Stop loss – this is again at a portfolio level based on loss ratio (claims to premium) for a given period of time. For example, if the loss ratio exceeds 125%, then a proportion (say, 70%) of claims in excess of office premium will be paid by the reinsurer. There can also be an upper limit on the loss ratio beyond which the risk again falls back on the insurer.
- Catastrophe cover – this protects the insurer from a single event leading to a lot of claims. for example, an earth quake leading to a number of insured lives getting hospitalized. There are typically limits that apply such as minimum no. of lives to be affected and the maximum payable in one event or in one year.

[7]

ii)

- Risk appetite of the insurer – new risk with no own past experience; so, a higher level of reinsurance would be preferable to start with. While there may be standard hospital cash products selling in the market, the unique characteristic of increasing per day benefit adds to the riskiness of the product.
- If there is a lot of credible data on flat benefit hospital cash products in the market, then an individual surplus arrangement may be desirable.
- Otherwise, a flat quota share with higher proportion reinsured might be prudent until the insurer's own experience builds up.
- The product is claim intensive unlike life insurance products. So, the level of reliance on the reinsurer for claim management would also drive the type and level of arrangement. Similarly, the level of underwriting support needed would help.
- The reinsurer would expect a higher reinsurance in return for the technical support expected.
- Current level of free capital would also need to be factored in.
- The higher the free capital the greater the risk that the insurer can take on and so lower level of reinsurance required.
- Geographical spread of the lives to be covered (depends to some extent on the spread of distribution) would need to be looked at to determine the need for any non-proportional covers such as a catastrophe cover. The more well-spread the lives are, the lower the need for such a cover.
- Need for any pandemic cover should also be considered.
- Competitiveness of the reinsurance terms is another important factor.
- There could be significant capital and tax arbitrages making it attractive to buy more reinsurance
- Given that there is life time renewability and 5-year rate reviewability, it might be important to check if the reinsurance arrangements provide back-to-back guarantees.
- The expected size of business generated from this product relative to overall new business volumes is another factor to consider.
- The greater the share of this product in new business the larger the importance of reinsurance in order to manage overall riskiness of the portfolio.

[7]

[14 Marks]

Solution 5:

i)

The key focus for a regulator is protection of policyholders' interest. The regulator would therefore expect you to examine:

- Mis-selling - whether there have been instances of mis-selling
- and, what has been done address the issue.
- This product being first of its kind there is a possibility that the intermediaries have neither understood or unable to explain to the policyholders.
- You will need to look at the policy holder complaints and persistency of the product to identify if there is mis-selling.
- And, communicate to the regulator on the steps taken to control mis-selling and demonstrate how effective they have been.
- The steps might involve things such as making the sales communication more clear & call up the customers after the sale to check if they have understood the policy terms correctly.
- Fair deal – Is the policyholder getting a fair deal? Are the premiums appropriate to the level of benefit offered?
- You would need to revisit the pricing assumptions and profit-test to ensure the premiums are neither excessive nor insufficient. You might have done the exercise at the end of 3-year guarantee depending upon the level of credibility.
- If there was a premium review and you are not intending to make any further changes, you will need to demonstrate why there is no further review of premium required.
- Again, given that the product is new the pricing assumptions might have been overly conservative and so require adjustment as the experience emerges.
- Claim management – what proportion of claims have been repudiated? Higher proportion would indicate disconnect between what the policyholders' expect the policy to pay and what has actually been done.
- Need to examine whether all genuine claims are paid and you may get the reinsurer to audit claim processes to ensure the level of claim underwriting is appropriate.
- A significant disconnect might require revisiting the policy definition to make them simpler for the policyholder to understand/
- Business volume – expected vs actual – the regulator would expect you to have looked at the variance from numbers projected at the time of original filing.
- Any material deviation would need to be explained and a revised projection made including its impact on the pricing of the product submitted.

- Definition of critical illnesses – Any medical advances leading to a particular definition being no longer appropriate?
- You will need to consult the underwriters, claim managers and reinsurers to ensure that the definitions remain relevant.
- Compliance – you will need to examine and ensure any regulatory changes after the product was originally filed are complied with.
- Also, any particular area that the regulator had asked you to examine before the review after 5 years should also be looked at.

[7]

ii)

- It is a good idea and might help reduce over-use of insurance due to the tendency to use up to the maximum benefit.
- The policyholder may tend to stay in the hospital longer or resort to more expensive treatment options if the sum assured is too large relative to the level of treatment costs expected.
- It will make the product more complicated - One sum assured limit is simpler compared to multiple limits.
- As it is a new product and any further complication make it difficult for the intermediaries to explain and for the policyholders to understand.
- So, it might lead to reduction in sales.
- Differentiating the maximum benefit linked to the nature of critical illnesses may prove to be a new product feature in the market and thus help improve volumes and be more competitive.
- This could prove to be a good strategy if someone else has already copied your current product and you no longer enjoy the monopoly.
- The new feature will continue to help you differentiate from the rest of the market and potentially build a brand of being product innovator in the market.
- Other things being equal, the feature could lead to reductions in premium rates without much perceived reduction in the value of the product
- the product still meets reasonable expenses of treating a critical illness.
- The administrative system implications for offering such a feature will need to be considered
- ...as will any regulatory constraints.

[4]

[11 Marks]

Solution 6:

i)

- The key risk to the insurer is that of parameter mis-estimation – the no. of lives who would exercise the option
-and the level of health of those who exercise the option.
- The lower the no. of lives who exercise the option the higher the level of sub-standard amongst those who exercise
- ...and thus resulting in heavier morbidity experience than expected.
- The decision to exercise the option at the end of 10 years will depend on a number of factors - the individual's health status then
-(any family history, peer or friends suffering critical illness and any symptoms of critical illness etc. – lives in poor health would tend to exercise the option),
-the company's brand image with respect to critical illness product such as claim repudiation and price competitiveness
-(healthy lives who want to have the critical illness production may go elsewhere)
- Increased awareness on critical illness (for example, government campaigns, press coverage on critical illness trends etc.) might lead to higher take up.
- While there is no fresh underwriting, a waiting period of six months may be applied at the end of 10 years during which CI claims will not be payable in order to avoid those claims-in-waiting.
- The option may be offered to those buying term policies up to a sum assured (say, up to 50 lacs) to avoid over-insurance for CI and thus creating wind-fall opportunities.
- Alternatively, the sum assured under CI can be limited to a portion of the term life sum assured up to max 50 lacs.
- The term of CI cover can be reduced to 10 years from the 20 year period – so that the policy term for both life and CI are equal.
- A lien? that can be introduced such that a small cover is payable in the initial years and it increases to full cover gradually over time.
- Buy reinsurance protection to limit the exposure to risk
- and/or apply limit on business volumes so that any significant underpricing does not threaten the solvency of the company
- Introduce well-ness related incentives encouraging policyholders to monitor and maintain their health
-so that the pool's health overall improves

[7]

ii)

The assumed premium from age 35 for the policy taken out under the option (the “option policy”) is:

$$50,00,000 \frac{A_{[45]}^1}{\ddot{a}_{[45]:20}} = 50,00,000 \times \frac{0.0589}{13.785} = 21,377.95 \text{ (or 21363.80, if figs taken up to 4 digits)}$$

$$\begin{aligned}
 (A_{[45]}^1 :_{20} &= A_{[45]} - A_{65} \times v^{20} \frac{l_{65}}{l_{[45]}} = 0.27583 - 0.52786 \times \frac{8821.2612}{9798.0837} \times (1.04)^{(-20)} \\
 &= 0.27583 - 0.52786 \times 0.4109 = 0.0589)
 \end{aligned}$$

The expected present value at age 25 of the cost of the option is:

$$\begin{aligned}
 &(50,000 A_{45}^1 :_{20} - 21,377.95 \ddot{a}_{45:20}) v^{10} \frac{l_{35}}{l_{[25]}} \\
 &= (50,000 \times 0.0592 - 21377.95 \times 13.780) \times (1.04)^{-10} \times \frac{9894.4299}{9951.9913} \\
 &= \text{INR } 948.28 \text{ (or 1050.55, if exact figs are taken)}
 \end{aligned}$$

$$\begin{aligned}
 (A_{45}^1 :_{20} &= A_{45} - A_{65} \times v^{20} \frac{l_{65}}{l_{45}} = 0.27605 - 0.52786 \times \frac{8821.2612}{9801.3123} \times (1.04)^{(-20)} \\
 &= 0.27605 - 0.52786 \times 0.41075 = 0.0592)
 \end{aligned}$$

[5]

[12 Marks]

Solution 7:

i)

- Close matching minimizes investment risk by improving certainty around the ability to meet liability outgo as and when they fall due. This would therefore reduce the risk of insolvency
- Realizing assets at an inopportune time might lead to very poor investment returns and that could threaten the solvency
- Close matching in terms of timing and nature helps minimize this risk.
- However, close matching per se does not lead to better investment returns.
- The returns depend on the nature of the assets and their associated riskiness.
- Some companies do deliberate mis-matching in order to achieve higher returns. This can be done when there are a lot of free assets and adequate liquidity.
- Further, the matching at product level would prove to be an ineffective investment approach if there is not enough scale
- So, it is ok to look at the portfolio level to try and match the cash-flows as closely as possible.

[3]

ii)

- LTCI liabilities would normally be real, with claims related directly or indirectly to future care costs.
- The assets that would be required should give income which is fairly stable and increasing in real terms

- The liabilities are expected to be fairly long term and short term volatility should be less of a concern;
- Therefore investments in equity, index linked bonds and long dated bonds could be considered
- The markets could still lead to losses and the long term bonds would also be volatile due to interest rate movements.
- However, over the long term they would be expected to provide a better match to the nature of liabilities of this product
- It is also suggested that the company should switch from equities to say index linked bonds
-as the liabilities become more mature, to reduce the extent of risk

[3]

[6 Marks]
